Block Quest -- where the cubes go

You need the six tagged cubes (tags 0 to 5, the 40 mm AprilTags from apriltag_16h5_40mm.pdf, one on each cube's faces and top) and the frame cube (tags 24 to 29). The game shows the same layout on the video as ghost footprints and a checklist at the start of every level, and starts when the cubes are in place -- this sheet is the same plan on paper. All positions are measured from the frame cube's centre: 'beyond' is away from the robot, left and right as the robot sees them.

1. The Dawn Stair -- puzzle · climbing
2. The Cinder Causeway -- lava · platforms · move a block
3. The Sentinels' Court -- melee · shield · archers
4. The Powder Cellar -- bombs · armor · powder kegs
5. The Sunbeam Vault -- puzzle · mirrors · no enemies
6. The Ember Ferry -- co-op · two players · lava river
7. The Bone Warden -- boss · sword · shield · bombs · kegs

Level 1 -- The Dawn Stair

puzzle · climbing (cube size 5 cm; the game scales with the measured size)

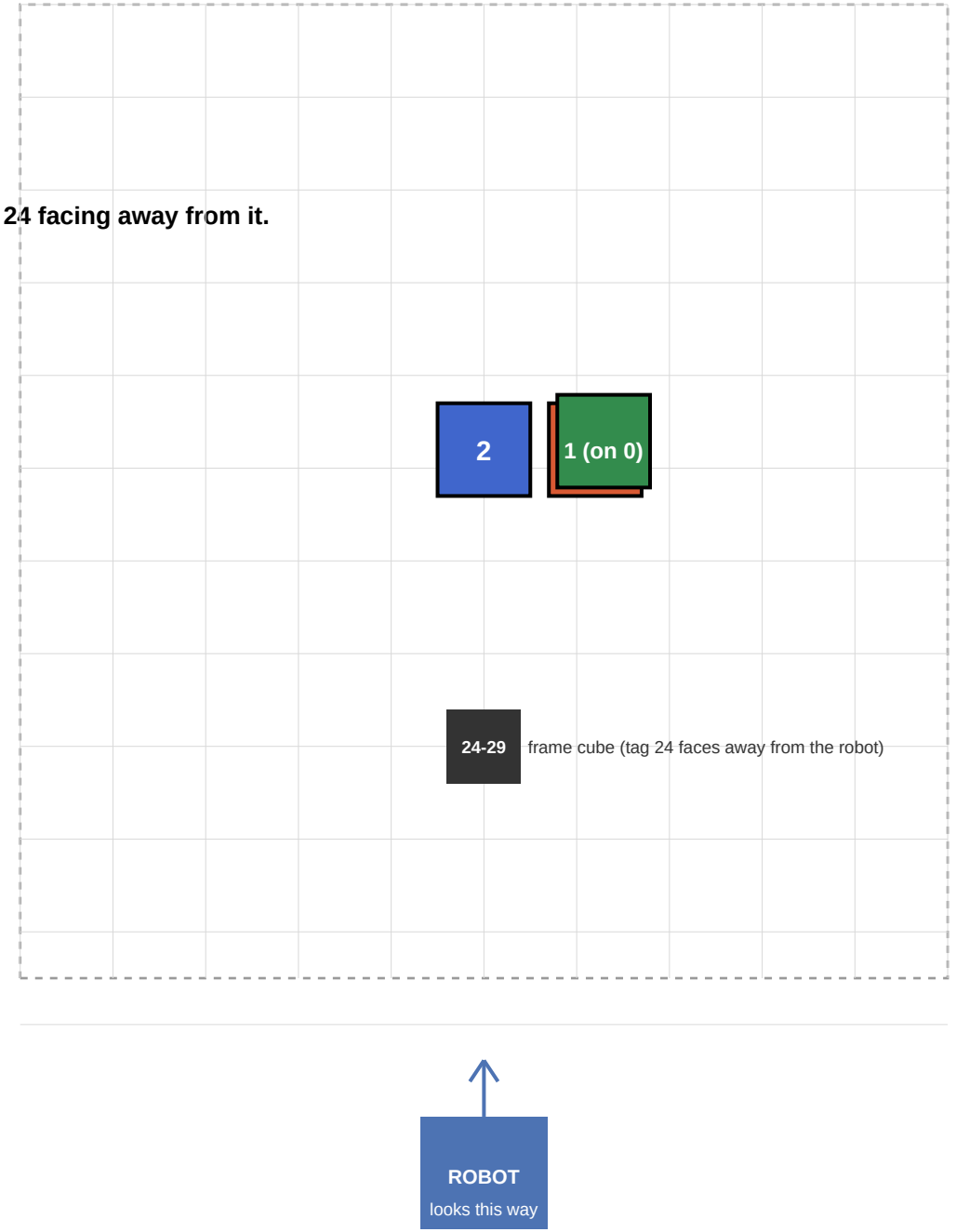
The Sunstone lies on the top of a ruined tower. Climb it, take the stone, and carry it back to the beacon (the frame cube) to bring the dawn.

Frame cube (tags 24-29): the origin. About 25 cm in front of the robot, tag 24 facing away from it.

Cube 0 Tower: 16 cm beyond the frame cube, 6 cm to the right

Cube 1 Tower top: 16 cm beyond the frame cube, 6 cm to the right -- on top of cube 0

Cube 2 Step: 16 cm beyond the frame cube



Top view. Grid: one cube (5 cm). Dotted: the level's play area. Left of the page = left of the robot.

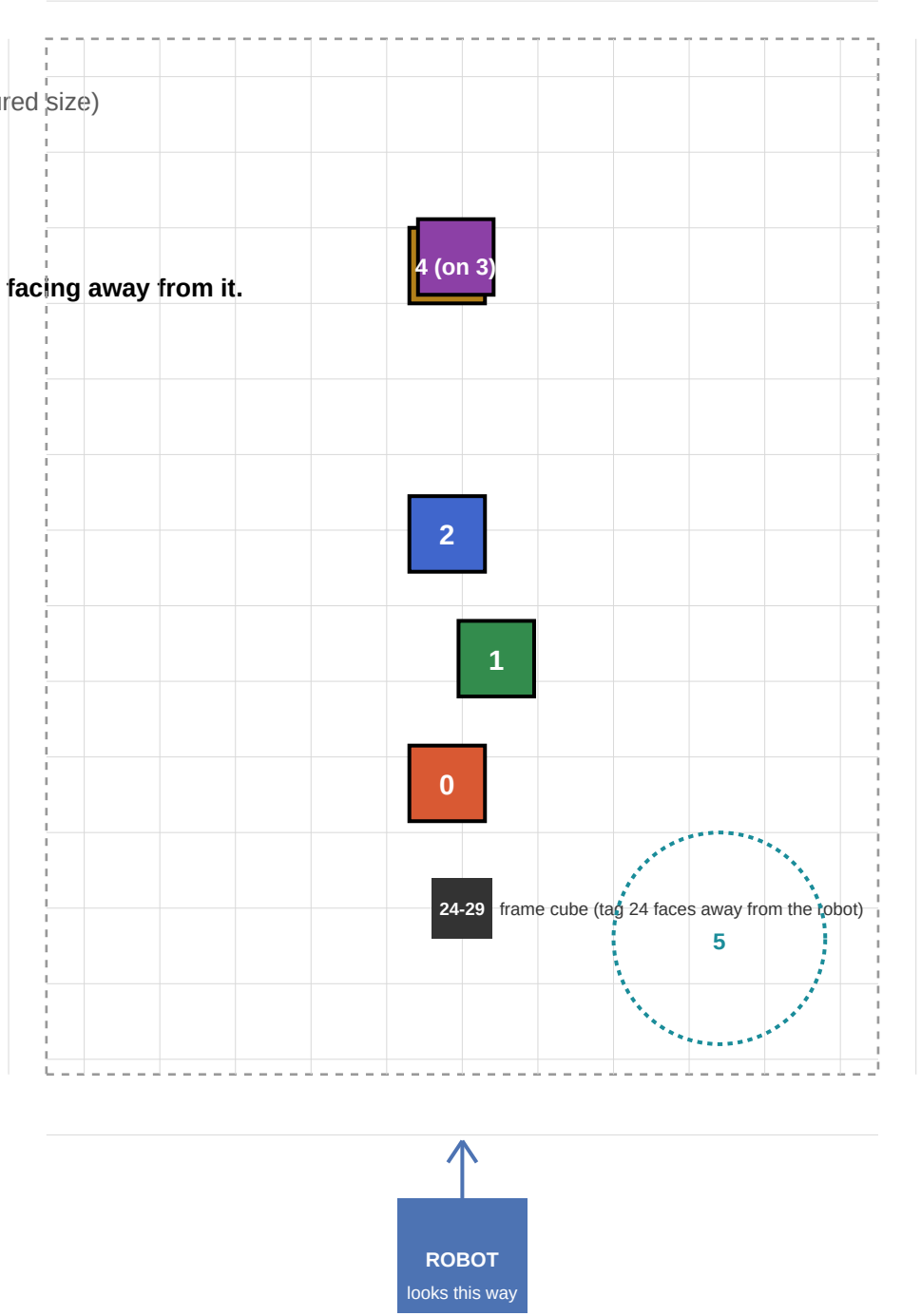
Level 2 -- The Cinder Causeway

lava · platforms · move a block (cube size 5 cm; the game scales with the measured size)

The temple floor has melted. The only way to the Ember Crown is over the stones. One gap is too wide for any jump: carry the spare stone to it -- with your hands, or the robot -- but never one the knight stands on.

Frame cube (tags 24-29): the origin. About 25 cm in front of the robot, tag 24 facing away from it.

Cube 0 Stone 0: 8 cm beyond the frame cube, 1 cm to the left
Cube 1 Stone 1: 16 cm beyond the frame cube, 2 cm to the right
Cube 2 Stone 2: 25 cm beyond the frame cube, 1 cm to the left
Cube 3 Pedestal: 42 cm beyond the frame cube, 1 cm to the left
Cube 4 Pedestal top: 42 cm beyond the frame cube, 1 cm to the left -- on top of cube 3
Cube 5 Spare stone: 2 cm short of the frame cube, 17 cm to the right (anywhere in the ring)



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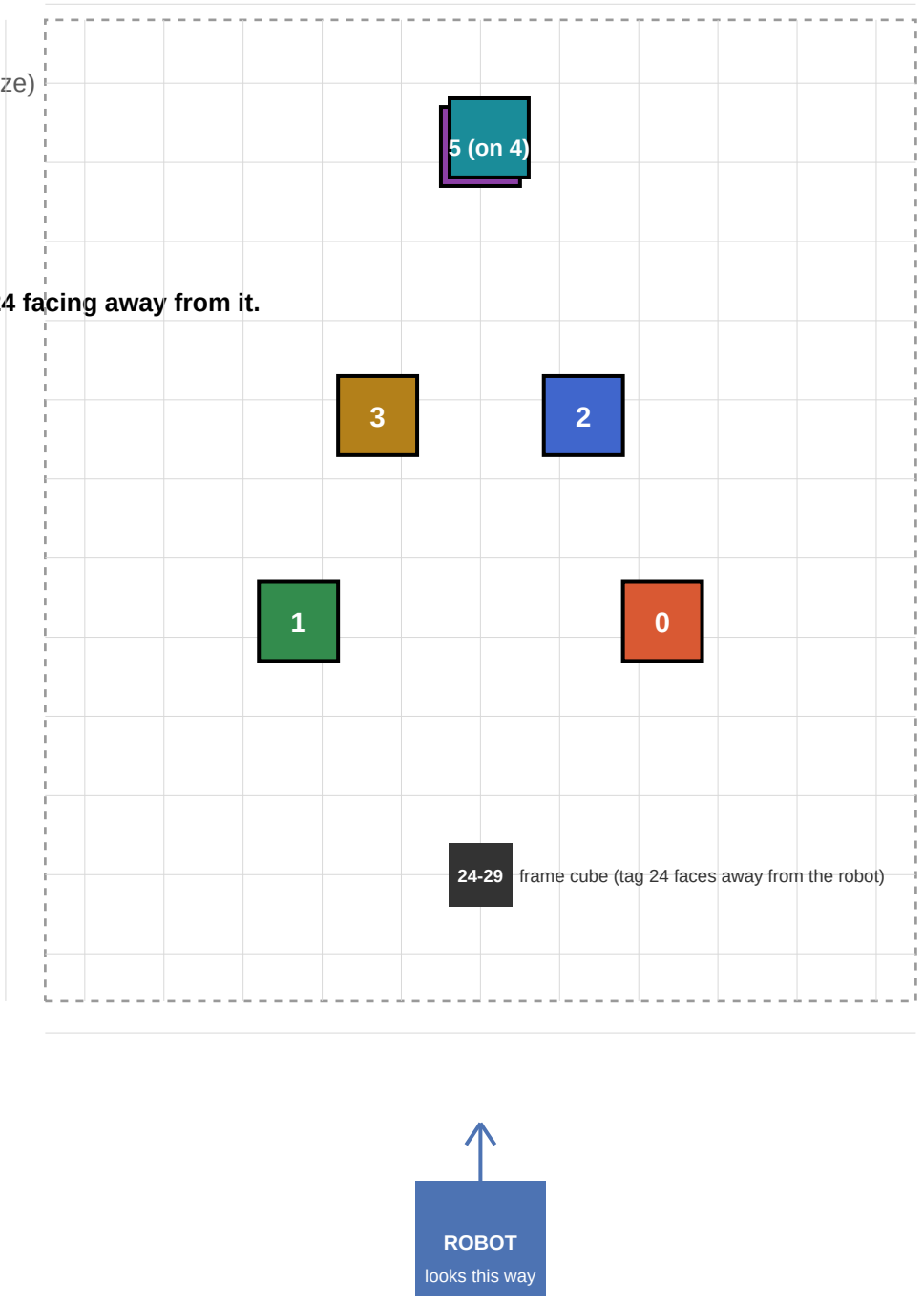
Level 3 -- The Sentinels' Court

melee · shield · archers (cube size 5 cm; the game scales with the measured size)

The court is guarded by the dead. Their blows are slow, and tell you when they come: raise your shield just as one lands to parry it, then strike. Hoplites hide behind their own shields; the archer on the tower shoots over the pillars you leave him.

Frame cube (tags 24-29): the origin. About 25 cm in front of the robot, tag 24 facing away from it.

Cube 0 Pillar: 16 cm beyond the frame cube, 12 cm to the right
Cube 1 Pillar: 16 cm beyond the frame cube, 12 cm to the left
Cube 2 Pillar: 29 cm beyond the frame cube, 6 cm to the right
Cube 3 Pillar: 29 cm beyond the frame cube, 6 cm to the left
Cube 4 Tower base: 46 cm beyond the frame cube
Cube 5 Tower top: 46 cm beyond the frame cube -- on top of cube 4



Top view. Grid: one cube (5 cm). Dotted: the level's play area. Left of the page = left of the robot.

Level 4 -- The Powder Cellar

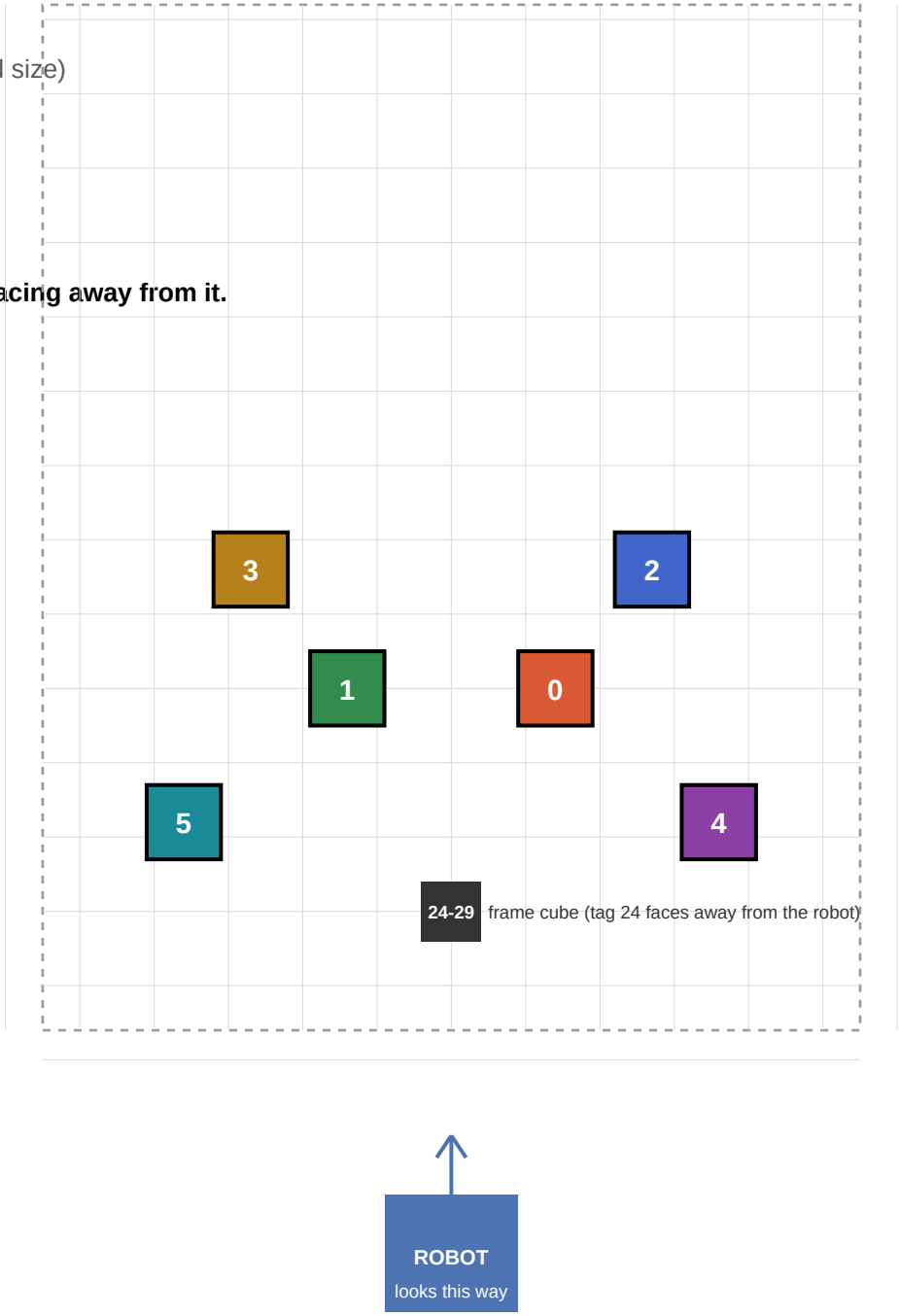
bombs · armor · powder kegs (cube size 5 cm; the game scales with the measured size)

The vault's door is a heap of rubble that only powder can move, and the brute behind it is armored: the sword rings off its plates until a blast cracks them. Bombs go where you set them and wait; the kegs (cubes 4 and 5) go where you put them -- by hand.

Frame cube (tags 24-29): the origin. About 25 cm in front of the robot, tag 24 facing away from it.

Cube 0 Pillar: 15 cm beyond the frame cube, 7 cm to the right
Cube 1 Pillar: 15 cm beyond the frame cube, 7 cm to the left
Cube 2 Pillar: 23 cm beyond the frame cube, 14 cm to the right
Cube 3 Pillar: 23 cm beyond the frame cube, 14 cm to the left
Cube 4 Powder keg: 6 cm beyond the frame cube, 18 cm to the right
Cube 5 Powder keg: 6 cm beyond the frame cube, 18 cm to the left

You start with 3 bombs.



Top view. Grid: one cube (5 cm). Dotted: the level's play area. Left of the page = left of the robot.

Level 5 -- The Sunbeam Vault

puzzle · mirrors · no enemies (cube size 5 cm; the game scales with the measured size)

A sun-orb throws a beam across the floor. The cubes are mirrors: set one at 45 degrees and the beam turns a right angle. The prism (cube 3) splits a beam in two. The frame cube and the obsidian block swallow light. Light both crystals -- then the vault door lifts.

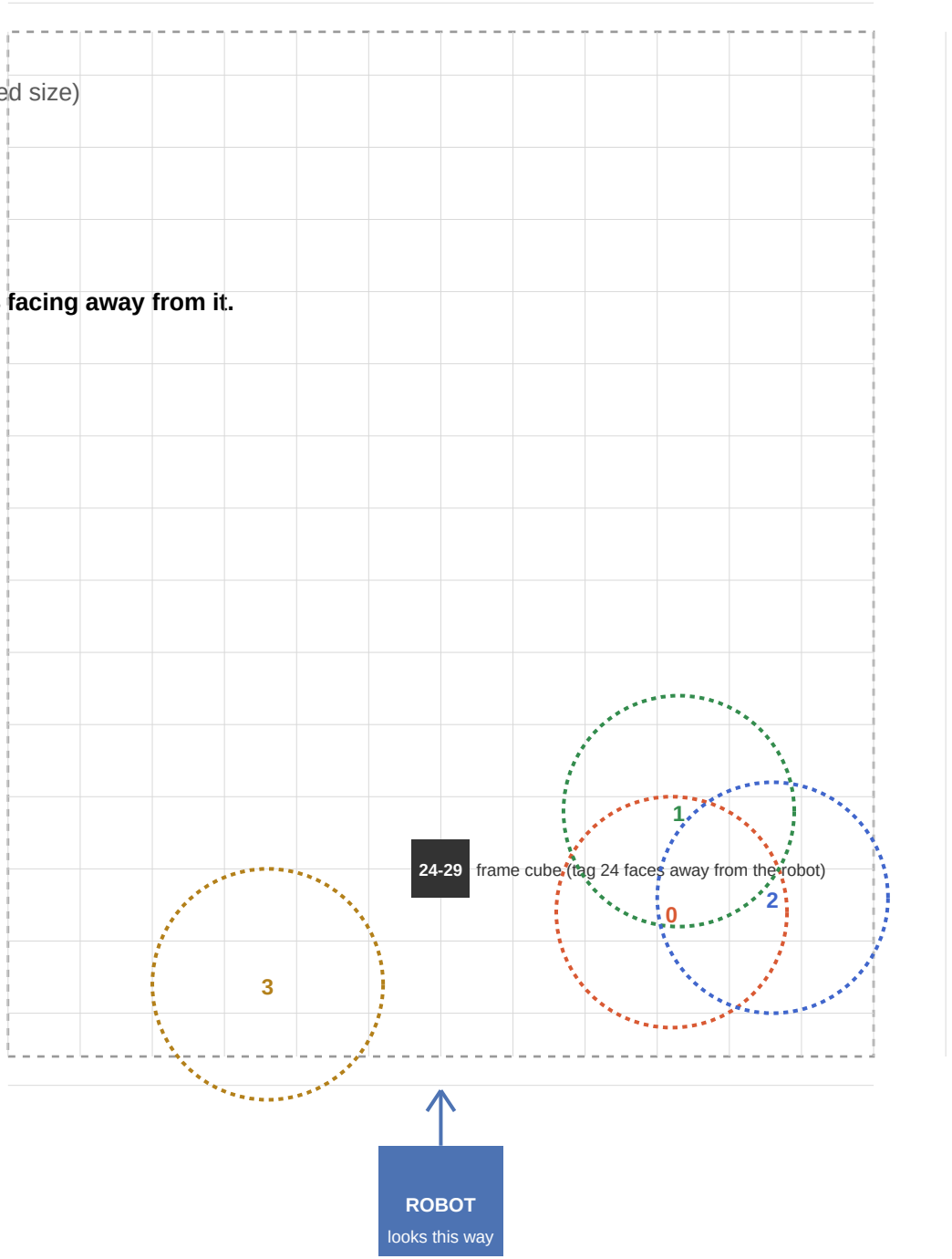
Frame cube (tags 24-29): the origin. About 25 cm in front of the robot, tag 24 facing away from it.

Cube 0 Mirror: 3 cm short of the frame cube, 16 cm to the right (anywhere in the ring)

Cube 1 Mirror: 4 cm beyond the frame cube, 16 cm to the right (anywhere in the ring)

Cube 2 Mirror: 2 cm short of the frame cube, 23 cm to the right (anywhere in the ring)

Cube 3 Prism: 8 cm short of the frame cube, 12 cm to the left (anywhere in the ring)



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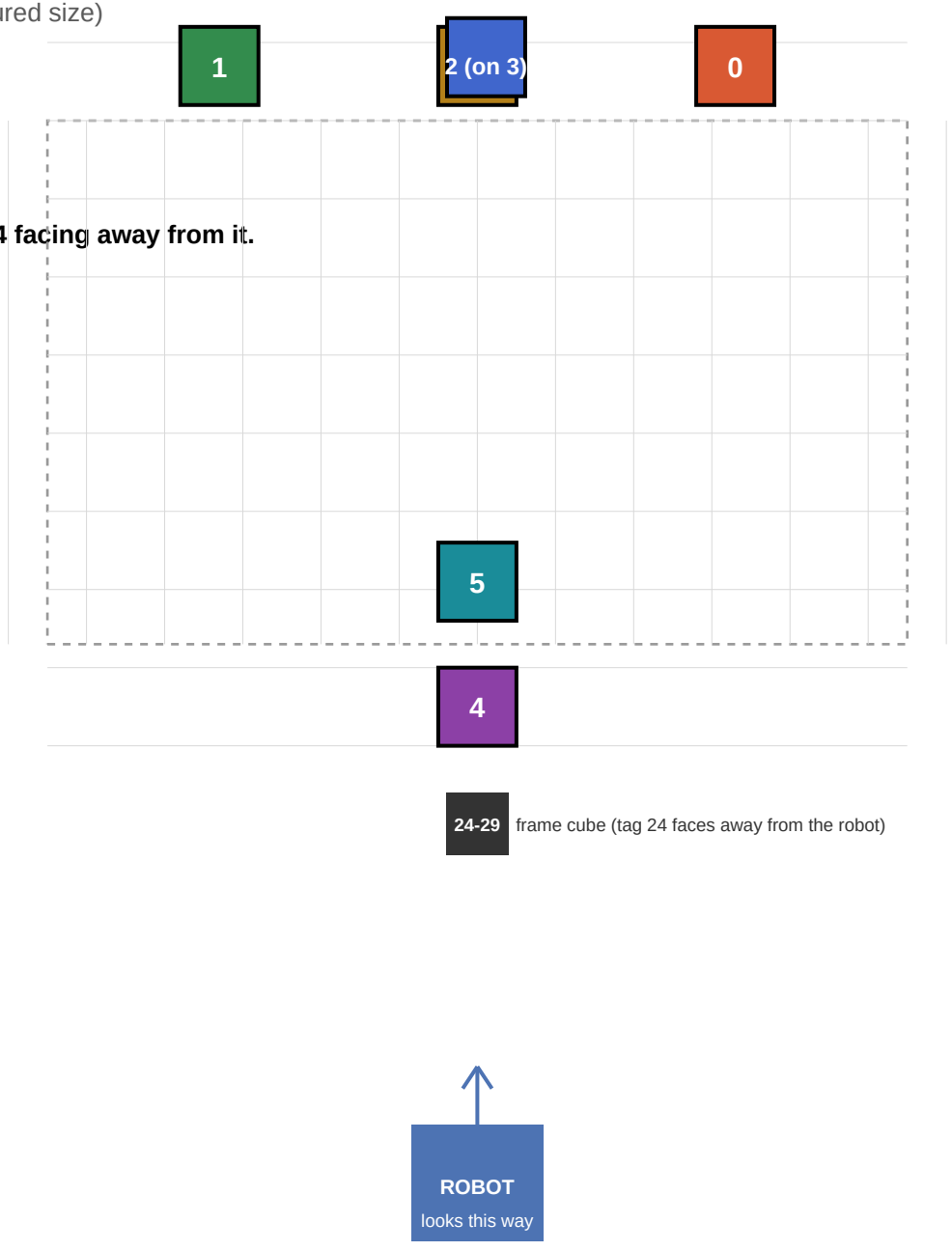
Level 6 -- The Ember Ferry

co-op · two players · lava river (cube size 5 cm; the game scales with the measured size)

A river of lava between the docks, and a ferry that has no ferryman. Player 2 is the ferryman: slide cube 5 across by hand, round the grey rocks (a bump hurts the knight) and clear of the geysers, while the knight stands on it and shields itself from the archers on the far bank.

Frame cube (tags 24-29): the origin. About 25 cm in front of the robot, tag 24 facing away from it.

Cube 4 Near dock: 8 cm beyond the frame cube
Cube 5 THE FERRY (Player 2 moves it): 16 cm beyond the frame cube
Cube 3 Far dock: 48 cm beyond the frame cube
Cube 2 Dock top: 48 cm beyond the frame cube -- on top of cube 3
Cube 0 Archer's perch: 48 cm beyond the frame cube, 16 cm to the right
Cube 1 Archer's perch: 48 cm beyond the frame cube, 16 cm to the left



Top view. Grid: one cube (5 cm). Dotted: the level's play area. Left of the page = left of the robot.

Level 7 -- The Bone Warden

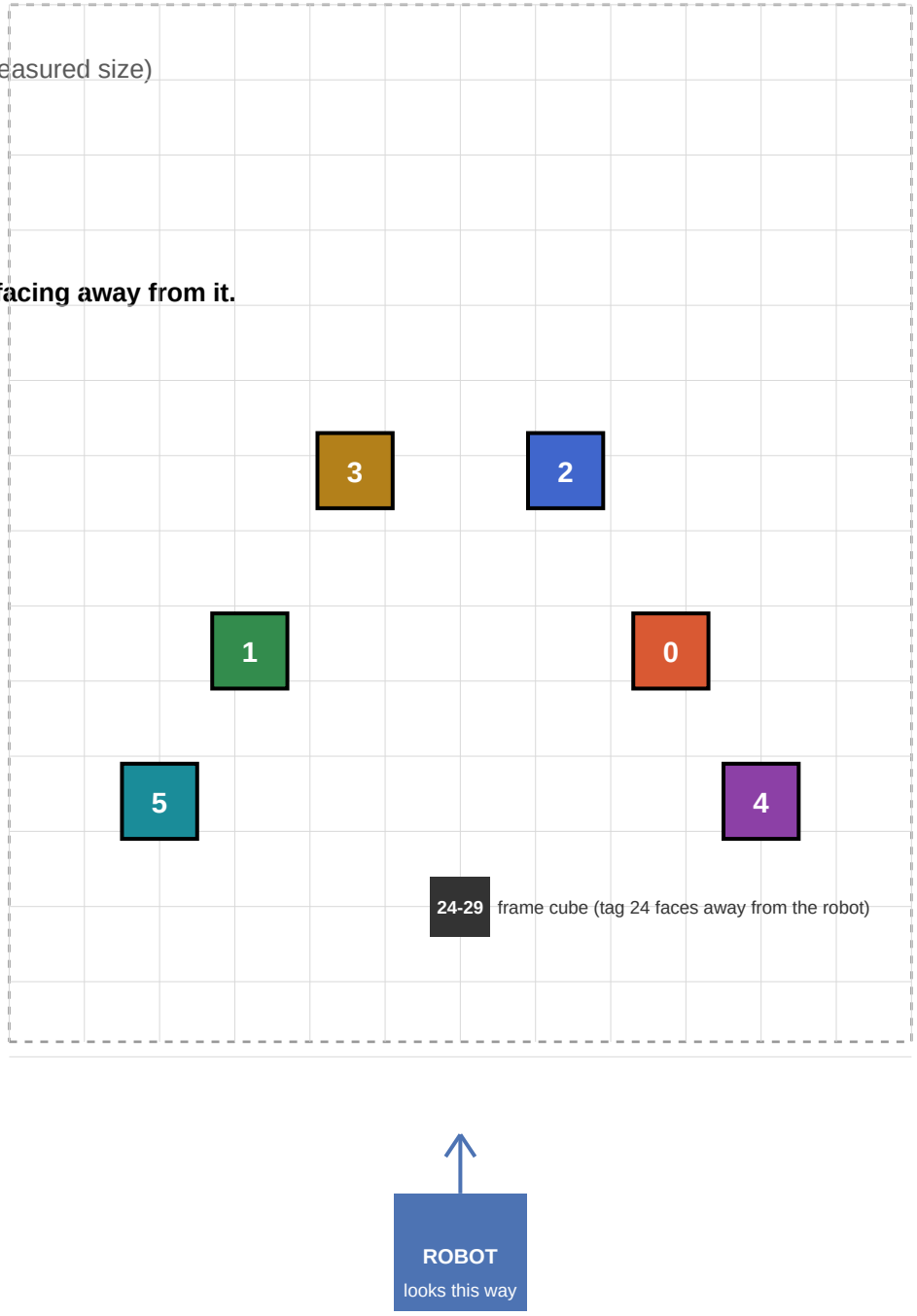
boss · sword · shield · bombs · kegs (cube size 5 cm; the game scales with the measured size)

The Warden has stood here since the temple was built. Its slam sends a ring along the floor -- jump it, or stand on a block. Its sweep can be parried. Its armor is only cracked by a blast: bombs, or a powder keg you move into its path -- then strike hard before the bones knit again.

Frame cube (tags 24-29): the origin. About 25 cm in front of the robot, tag 24 facing away from it.

Cube 0 Pillar: 17 cm beyond the frame cube, 14 cm to the right
Cube 1 Pillar: 17 cm beyond the frame cube, 14 cm to the left
Cube 2 Pillar: 29 cm beyond the frame cube, 7 cm to the right
Cube 3 Pillar: 29 cm beyond the frame cube, 7 cm to the left
Cube 4 Powder keg: 7 cm beyond the frame cube, 20 cm to the right
Cube 5 Powder keg: 7 cm beyond the frame cube, 20 cm to the left

You start with 4 bombs.



Top view. Grid: one cube (5 cm). Dotted: the level's play area. Left of the page = left of the robot.